

The Shared Representation Hub

How seven encoders, entry maps, the DINOv2-small route, and a BGE-M3 caption head fit together



The central idea

The system separates **representation alignment** from **task decoding**. Every encoder gets its own entry map into one shared 512-dimensional coordinate system. A reusable caption head then converts Hub coordinates into the BGE-M3 space where caption retrieval is evaluated.

Many private encoder languages → one shared internal language → one searchable caption language

Most important distinction

The original Hub $\mathbf{H}$ is built jointly from all participating encoders. $\mathbf{H}_{\text{small}}$ is only DINOv2-small's reconstruction of that Hub. They are related, but they are not the same matrix.

1. Building the frozen Hub

There are 9,533 aligned image-caption rows. The first 8,533 fitting rows build the Hub; 1,000 rows are held out for evaluation. Each row refers to the same underlying image-caption item across every encoder.

Each encoder matrix is standardized independently, then the seven matrices are concatenated column-wise.

$$\mathbf{C} = [\mathbf{Z}_{\text{img-small}} \mid \mathbf{Z}_{\text{img-base}} \mid \mathbf{Z}_{\text{img-large}} \mid \mathbf{Z}_{\text{BGE}} \mid \mathbf{Z}_{\text{GPT-2}} \mid \mathbf{Z}_{\text{BERT}} \mid \mathbf{Z}_{\text{SBERT}}]$$

PCA/SVD compresses this wide joint matrix to the top 512 directions. Whitening equalizes their scale. The result is:

$$\mathbf{H} \in \mathbb{R}^{8533 \times 512}$$

Part	Meaning
One row $\mathbf{h}_j$	The shared 512-d representation of training item j .
One column	One PCA-derived direction across all training items.
$H_{j,k}$	Item j 's coordinate on Hub direction k .

After fitting, the standardization statistics, PCA directions, whitening scales, and $\mathbf{H}$ are frozen. The held-out 1,000 rows do not alter them.

$\mathbf{H}$ depends on all seven encoders - not only on DINOv2-small.

2. One independent entry map per encoder

Encoder k produces raw vectors X_k with dimension d_k . Ridge regression fits a matrix W_k that predicts the already-frozen Hub:

$$X_k W_k \approx H, \quad W_k \in \mathbb{R}^{d_k \times 512}$$

Each map is fitted independently. W_{small} sees only X_{small} and H ; it never sees another encoder's map. The same rule applies to every encoder.

DINOv2-small example

X_{small} has 768 columns. Its entry map has shape 768 x 512:

$$H_{\text{small}} = X_{\text{small}} W_{\text{small}} \approx H$$

H_{small} is DINOv2-small's estimate of the Hub. The approximation is imperfect:

$$H_{\text{small}} = H + E_{\text{small}}$$

E_{small} is the entry-map reconstruction error. Other encoders produce H_{base} , H_{large} , H_{BGE} , and so on, each with its own error pattern.

Why ridge?

The regularization term stabilizes the solution when source dimensions are correlated or weakly supported. The map is solved through matrix algebra rather than epochs of neural-network training.

Entry map = translator from an encoder's private coordinates into the common Hub coordinates.

3. The DINOv2-small-trained caption head

The phrase can be misleading: DINOv2 and BGE weights are **not merged**. The caption head is a separate 512×1024 linear matrix trained from paired examples.



For each training row

The input is the image-derived Hub vector $h_{\text{small},j}$. The desired output is the BGE-M3 vector $t_{\text{BGE},j}$ of the matching caption.

$$H_{\text{small}} W_{\text{head}} \approx T_{\text{BGE}}, \quad W_{\text{head}} \in \mathbb{R}^{512 \times 1024}$$

Ridge regression chooses W_{head} so that, across all paired rows, each image-derived Hub vector is mapped near its matching caption's BGE vector. It learns cross-modal correspondence from row alignment.

Why use H_{small} instead of perfect H ?

At deployment the head receives coordinates produced through the DINOv2-small entry route, including its mapping errors. Training on H_{small} matches this real input distribution.

Why is the head necessary?

The Hub is a new 512-d PCA coordinate system. Even though BGE helped build it, the Hub is not BGE space. Candidate captions live in BGE's 1024-d space, so the head is the bridge that makes cosine retrieval possible.

The head predicts a BGE vector. It does not generate caption words.

4. Image-to-text and text-to-image retrieval

For an image, the complete frozen route is:

image $\rightarrow$ DINO-small $\rightarrow$ W_{small} $\rightarrow$ H_{small} $\rightarrow$ W_{head} $\rightarrow$ predicted BGE vector

Image to caption

Compare the predicted BGE image vector with stored BGE caption vectors. The nearest caption by cosine similarity is retrieved.

Text to image - no inverse head is required

Encode the user's text query directly with BGE. Separately, precompute the predicted BGE vectors for all database images using the route above. Compare the query vector against those image-derived BGE vectors and retrieve the nearest image.

BGE(text query) $\leftrightarrow$ predicted-BGE(database images)

If the goal is specifically to predict a raw DINO vector

That is a different experiment. Fit a new image exit head W_{image} : $512 \rightarrow 768$, for example $H_{\text{BGE}} W_{\text{image}} \approx X_{\text{small}}$. For normal text-to-image retrieval, staying in BGE space is simpler and avoids this extra decoder.

Cosine similarity is symmetric, so one common BGE comparison space supports both retrieval directions.

5. Adding an unseen encoder

To connect ConvNeXt, SigLIP, or another new encoder, keep the Hub and caption head frozen:

1. Cache the new encoder's vectors on the aligned fitting rows.
2. Fit only $X_{\text{new}} W_{\text{new}} \approx H$.
3. Apply the unchanged W_{head} to $H_{\text{new}} = X_{\text{new}} W_{\text{new}}$.
4. Evaluate in BGE space on held-out rows.

$$X_{\text{new}} \rightarrow W_{\text{new}} \rightarrow H_{\text{new}} \rightarrow \text{frozen } W_{\text{head}} \rightarrow \text{BGE space}$$

What strong transfer means

If an unseen architecture performs well after fitting only its entry map, its Hub reconstruction is compatible with the coordinate convention expected by the frozen head. This supports the Hub as a reusable interface and provides evidence for recoverable shared structure.

What it does not prove

It does not prove that the Hub is a true universal or Platonic representation. PCA captures dominant joint variance, which can include shared information, encoder-specific information, or both. Stronger evidence comes from multiple unseen families, strict held-out evaluation, frozen hyperparameters, and direct-map baselines.

Complete system

$$\hat{T}_k = X_k W_k W_{\text{head}}$$

Object	Shape	Role
H	8533 x 512	Joint frozen Hub built from all encoders
H_{small}	8533 x 512	DINO-small reconstruction of H
W_{small}	768 x 512	DINO-small entry map
W_{head}	512 x 1024	Hub-to-BGE caption-space decoder